

Sushan Adhikari

AI/ML Engineer | NLP | Computer Vision | MLOps

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TECHNICAL SKILLS

Languages & ML:	Python; PyTorch, Scikit-Learn, OpenCV, YOLOv8; Pandas, NumPy
LLMs & RAG:	Fine-tuning (LoRA/QLoRA), RAG pipelines, hybrid search, cross-encoder rerankers, chunking
Inference & CV:	NVIDIA DeepStream, Triton Inference Server, GStreamer, TensorRT
MLOps & DevSecOps:	Docker, AWS/EC2, Terraform, CI/CD, SAST/DAST (DefectDojo, CodeRabbit), Git
Backend & Data:	FastAPI, Flask, REST APIs, PostgreSQL, Redis, Spring Boot (integration)
Other:	Monorepo architecture, CODEOWNERS, $\LaTeX$

EXPERIENCE

- **Junior MLOps Developer** — KyraWorks Jan – Apr 2026
 - Architected company-wide monorepo with CODEOWNERS governance; enforced branch protection and review policies across teams.
 - Integrated SAST/DAST security pipelines (DefectDojo, CodeRabbit) into CI/CD; triaged and resolved vulnerability findings.
 - Containerized services with Docker; provisioned and managed AWS EC2 infrastructure via Terraform.
- **Computer Vision Intern** — KyraWorks Oct 2025 – Jan 2026
 - Built and optimized real-time multi-camera inference pipelines using NVIDIA DeepStream and Triton Inference Server.
 - Benchmarked throughput across 10+ simultaneous streams; identified and resolved latency bottlenecks for production deployment.
- **Full-Stack Development Intern** — National Pension Management System, Nepal Jun – Oct 2025
 - Developed and deployed liveness detection module (Amazon Rekognition) for anti-spoofing on a platform serving 300k+ users.
 - Contributed to database schema design and backend API development (Spring Boot, PostgreSQL).
- **AI Data Volunteer** — [Mercuri.world](#) (Global Non-profit) Jan 2025 – Present
 - Designed job taxonomy and contributed to ML pipeline planning for a mental-health employment platform; sole Nepali member among 150+ global volunteers.

PROJECTS & STARTUPS

- **Nurvexa** — Co-founder nurvexalabs.com 2025 – Present
 - Designed and built multi-stage RAG pipeline with hybrid search, cross-encoder rerankers, and adaptive chunking for medical exam Q&A.
 - Engineered full production stack: Redis caching, PostgreSQL, Dockerized microservices, VPS hosting, and CI/CD deployment.
- **Neural Networks from Scratch** In Progress
 - Implementing MLP, backpropagation, SGD/Adam, CNNs, batch norm, dropout, RNNs, and attention from first principles in NumPy; trained on MNIST, CIFAR-10.
- **LLM from Scratch** In Progress
 - Building decoder-only transformer end-to-end: BPE tokenizer, multi-head self-attention, KV cache, positional encodings, and pre-training loop.
- **Statistics & Probability for Machine Learning** In Progress
 - Hands-on implementations covering MLE/MAP, Bayesian inference, Gaussian processes, information theory (KL divergence, mutual information), and concentration inequalities.

EDUCATION

B.Eng. Computer Engineering — Kathmandu University, School of Engineering Jan 2022 – Feb 2026
CGPA: 3.90/4.0

Exchange Semester, B.Tech CSE — Indian Institute of Technology Palakkad Jan – May 2025

SELECTED PUBLICATIONS

- **Directional Asymmetry in Low-Resource Legal Machine Translation: A Nepali–English Case Study**
Sushan Adhikari, S. Sharma, D. Lamichhane, R. Chulyadyo, B. K. Bal ACM ICAIL 2026
[\[manuscript\]](#) · [\[GitHub\]](#)

- **Enhancing Ethical Reasoning in Tiny LLMs via Fine-Tuning and Multi-Agent Consensus**
Sushan Adhikari, S. Sharma, D. Lamichhane, S. Sigdel *KUSET Vol.18 No.2, Dec 2024*
[\[manuscript\]](#) · [\[GitHub\]](#)
- **RAG for Theoretical CS Education: Algorithm Analysis & Complexity Theory**
Sushan Adhikari, S. Sharma, D. Lamichhane, U. Adhikari *IOP Conf. Series (USC 2025), In Press*
[\[submitted manuscript\]](#) · [\[GitHub\]](#)

ACHIEVEMENTS

- **Hult Prize OnCampus Winner** (2024) — CV-based fish-disease detection; represented Nepal at Regional Summit, Bangkok, Thailand.
- **Finalist** — ICT Awards 2024, Rising Star Innovation Category.
- **Winner, Health Track** — AI Crusade 2023 (first AI hackathon in Nepal).
- **Mahatma Gandhi Scholarship** — Indian Embassy, Nepal (2019–2021).

CERTIFICATIONS

[Machine Learning Specialization](#) — Andrew Ng (Stanford/Coursera, 2026) · [Embeddings for NLP](#) (OpenHPI, 2025)

LANGUAGES

Nepali (Native) **English** C1 Advanced — IELTS 8.0